STS MIDTERM STUDY NOTES

Indigenous Science → The Good Life

Based on the provided Science, Technology, and Society module.

LESSON 4: INDIGENOUS SCIENCE AND TECHNOLOGY IN THE PHILIPPINES

Indigenous Knowledge System

- Indigenous knowledge is learned and practiced within an indigenous community.
- It is embedded in daily life experiences and passed down by parents, elders, and community members.
- It is closely connected to culture, environment, values, stories, poems, and songs.

Examples of Indigenous Knowledge

- Predicting weather and seasons by observing animal behavior and celestial bodies.
- Using herbal medicine.
- Preserving food.
- Classifying plants and animals according to cultural properties.
- Selecting and preserving good seeds for planting.
- Using indigenous technology in daily life.
- Building local irrigation systems.
- Classifying soil types for planting.
- Producing wines and juices from tropical fruits.
- Growing plants and vegetables in household yards.

Indigenous Science

- Indigenous science is part of the Indigenous Knowledge System.
- It is practiced by different indigenous groups and early civilizations.
- It includes knowledge, expertise, practices, and representations.
- It guides interaction with the natural environment, especially in agriculture, medicine, understanding natural phenomena, and coping with environmental change.
- Science is part of culture; how science is practiced can depend on cultural practices.

Scientific Attitudes

- Motivating attitudes
- Cooperating attitudes
- Practical attitudes
- Reflective attitudes

Contributions of Indigenous Science

- Astronomy
- Pharmacology
- Food technology
- Metallurgy

Indigenous Science Process Skills

- Observing
- Comparing
- Classifying
- Measuring
- Problem solving
- Inferring

- Communicating
- Predicting

Indigenous Community Values

- Land is a source of life.
- Earth is respected as Mother Earth.
- Living and nonliving things are interconnected and interdependent.
- Humans are stewards/trustees of natural resources and have a responsibility to preserve them.
- Nature should be respected and properly cared for.

Why Indigenous Science Matters

- It contributed to the development of science and technology in the Philippines.
- It helped people understand the natural environment and cope with everyday life.
- UNESCO recognized indigenous science as a historical and valuable contribution to science and technology.

LESSON 1: HUMAN FLOURISHING

Eudaimonia

- Eudaimonia literally means "good spirited."
- It is associated with Aristotle (385–323 BC).
- It is commonly translated as human flourishing.
- It represents the pinnacle of happiness attainable by humans.

Aristotle’s Human Flourishing

- Phronesis
- Friendship
- Wealth
- Power

Human Flourishing Today

- Human flourishing changes as society changes.
- People pursue greater comfort, exploration, products, wealth, science, and technology.
- Modern humans are expected to become a 'man of the world' and work with institutions and government toward common goals.
- Coordination becomes increasingly important in a global society.

Eastern vs Western Views

- Western perspective: more individual-focused.
- Eastern perspective: more community-centered.
- Examples of community-centered traditions in the module include Chinese Confucianism and Japanese Bushido.
- Aristotle’s eudaimonia is presented as the ultimate good and may lead a flourishing person to serve the community through values and deliberation.

SCIENCE, TECHNOLOGY, AND HUMAN FLOURISHING

- Science and technology contribute to human knowledge and development.
- Humans use science to understand existence, the universe, and their role in the world.
- Science and technology can serve as tools toward human flourishing.
- Human flourishing should be pursued holistically, not through science and technology alone.
- A flourishing person develops intellectual, linguistic, kinetic, artistic, and socio-civic dimensions.

Science as Method

- Observe unexplained occurrences.
- Identify the problem and factors involved.

- Formulate a hypothesis.
- Conduct an experiment.
- Gather and analyze results.
- Formulate conclusions and recommendations.

Verification Theory

- A discipline is considered science when claims can be confirmed through evidence.
- It emphasizes empiricism, measurable results, and repeatable experiments.

Technology and the Good Life

- Technology can make life easier, more comfortable, and more efficient.
- Technology should not automatically be treated as the definition of the Good Life.
- Excessive reliance on technology can make people lose sight of what truly matters.
- People may begin to view nature and surroundings mainly according to economic value.

Backtracking the Human Condition

- Technology has contributed to longer lives, assistance for people with disabilities, industrial efficiency, scientific discoveries, and space exploration.
- Despite technological progress, poverty, conflict, resource competition, and the search for meaning still exist.
- Technological progress does not automatically equal human flourishing.

LESSON 3: THE GOOD LIFE

Aristotle and the Good Life

- Aristotle connected the goal of life with happiness.
- Humans move from potentiality toward actuality.
- Human actions are directed toward an end or purpose (telos).
- Eudaimonia is associated with human flourishing and the highest form of happiness.

Important Terms

- Telos → purpose or end.
- Potentiality → what something can become.
- Actuality → what something has become.
- Eudaimonia → human flourishing / highest form of happiness.

Happiness as the Goal of a Good Life — John Stuart Mill

- Greatest Happiness Principle: an action is right when it maximizes happiness for the greatest number.
- Ethical decisions consider benefits, harm, advantages, disadvantages, and overall happiness.

Schools of Thought About the Good Life

- Materialism → Emphasizes matter and material wealth as sources of happiness or meaning.
- Hedonism → Views pleasure as the goal of life; associated in the module with Epicurus.
- Stoicism → Emphasizes recognizing what is and is not within our control; associated with apatheia.
- Theism → Connects meaning and happiness with God and communion with God.
- Humanism → Emphasizes human freedom and the ability to shape one’s own destiny.

SUPER QUICK MEMORIZATION

- **Indigenous Science = Culture + Nature + Traditional Knowledge + Scientific Skills**
- **Indigenous Values = Land + Mother Earth + Interconnection + Stewardship + Respect**
- **Eudaimonia = Human flourishing / highest attainable happiness**
- **Aristotle = Happiness + Purpose + Human Flourishing**
- **Telos = Purpose / End**
- **Technology = Useful tool, but not automatically the Good Life**

- **John Stuart Mill = Greatest Happiness Principle**
- **Materialism = Wealth / Matter**
- **Hedonism = Pleasure**
- **Stoicism = Control what you can**
- **Theism = God**
- **Humanism = Human freedom and self-determination**

HIGH-PRIORITY EXAM TOPICS

- Definition and examples of Indigenous Science.
- Indigenous Science process skills.
- Four scientific attitudes.
- Indigenous community values.
- Eudaimonia and Aristotle’s concept of human flourishing.
- Eastern vs Western views of flourishing.
- Science and Technology as tools for human flourishing.
- Potentiality → Actuality and Telos.
- John Stuart Mill’s Greatest Happiness Principle.
- Materialism, Hedonism, Stoicism, Theism, and Humanism.
- Why technological progress does not automatically equal the Good Life.